

horizontal distance after the rebound, and land closer to the wall.

- 3 (a)** The mass is not uniformly distributed throughout the boat. The mass of the boat is more concentrated on the left side.

(b) 1. Zero resultant force acts on the boat, so that the boat is in translational equilibrium.

2. Zero resultant torque acts on the boat, so that the boat is in rotational equilibrium.

(c) $T_1 + T_2 = 15000 \text{ N}$

Taking moments about the point where rope 1 is attached to the boat,

$$(15000)(0.75) = T_2(2.00)$$

$$T_2 = 5625 \text{ N}$$

$$T_1 = 9375 \text{ N}$$

- 4 (a) (i)** Diffraction refers to the spreading of light waves when they pass through each of